Study design: a multi-tissue MRI aging clock and age acceleration

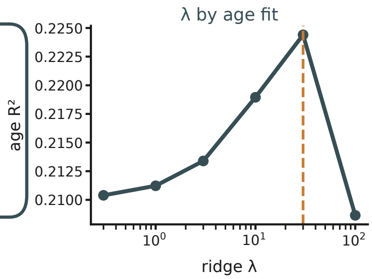
Cohort & segmentation

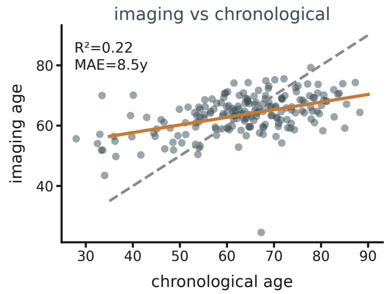
204 patients · preoperative lumbar MRI
Automated multi-tissue segmentation
(3D Slicer / TotalSegmentator)

Scanner-robust features:
size-normalized volumes + vertebra-referenced
T2 intensity ratios (iliopsoas, deep paraspinal,
vertebra, disc, cord)

MRI aging clock

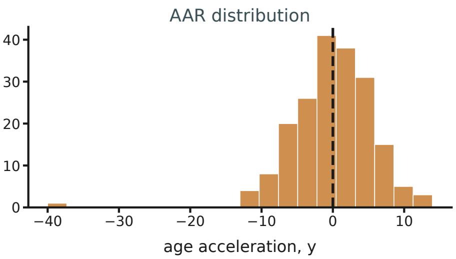
Ridge regression predicts
chronological age from the
multi-tissue signature
(10-fold CV; λ by age R²)





Age acceleration

Age acceleration (AAR) = residual of
imaging age ~ chronological age
Orthogonal to chronological age (no suppression)
positive = tissues look older than the patient's age



Association analysis

Firth penalized logistic regression

Non-home discharge ~ age acceleration
+ chronological age + sex + ASA class

Sensitivity: clock specification · λ · seed